

# Day 5 Plan

Maths: Number System · Reasoning: Coding-Decoding · Science: Physics 5:  
Waves & Sound

## Today's schedule

| # | Block           | Time   | What you do                                                                                 | Where       |
|---|-----------------|--------|---------------------------------------------------------------------------------------------|-------------|
| 0 | Day 4 recall    | 15 min | 10 questions from Day 4, no notes. Redo yesterday's mistakes.                               | Page 1-2    |
| 1 | Maths           | 75 min | Read the formulas & methods (10 min) → 2-3 book questions per pattern, timed → log mistakes | Section A   |
| - | Break           | 10 min | Away from the screen                                                                        | -           |
| 2 | Reasoning       | 50 min | Learn the method (10 min) → 15 timed questions from your book → log                         | Section B   |
| - | Break           | 10 min | Away from the screen                                                                        | -           |
| 3 | Science         | 50 min | Lessons with quick checks → 20-question practice set → answers                              | Section C   |
| 4 | Current affairs | 20 min | 20 MCQs from the monthly PDF                                                                | Your PDF    |
| 5 | Recall          | 10 min | Redo today's wrong questions; write 3 rules for tomorrow                                    | Mistake log |

Speed targets: Maths 60-70 sec per question · Reasoning 30-40 sec · Science 15-20 sec.

## Block 0 — Day 4 recall (answers on the last page)

| #  | Question                                                     | Your answer |
|----|--------------------------------------------------------------|-------------|
| 1  | SI unit of pressure?                                         |             |
| 2  | Why are dams thicker at the bottom?                          |             |
| 3  | Principle of the hydraulic lift?                             |             |
| 4  | Why does a ship rise a little entering the sea from a river? |             |
| 5  | Why are raindrops spherical?                                 |             |
| 6  | Principle behind aeroplane lift?                             |             |
| 7  | Which is more elastic, steel or rubber?                      |             |
| 8  | Which heat transfer needs no medium?                         |             |
| 9  | At what temperature is water densest?                        |             |
| 10 | Maths: CI – SI on ₹10,000 at 10% for 2 years?                |             |

Fewer than 8 right? Re-read only the lessons you missed — not the whole chapter.

# A · MATHS — Number System

## Core idea

Number system questions test **rules**, not long calculation. Learn the divisibility tests, unit-digit cycles and the factor formula below until you can apply each in under 10 seconds.

**Types of numbers:** natural (1, 2, 3 ...) · whole (0, 1, 2 ...) · integers (... -1, 0, 1 ...) · rational (p/q) · irrational ( $\sqrt{2}$ ,  $\pi$ ) · **prime** (exactly 2 factors; **2 is the only even prime; 1 is neither prime nor composite**; 25 primes below 100) · co-prime (HCF = 1, e.g. 8 and 15).

## Formulas

| Concept               | Formula / rule                                                                                                                    |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Divisible by 2, 5, 10 | Last digit even / 0 or 5 / 0                                                                                                      |
| Divisible by 3, 9     | Sum of digits divisible by 3 / by 9                                                                                               |
| Divisible by 4, 8     | Last 2 digits divisible by 4 / last 3 digits by 8                                                                                 |
| Divisible by 6        | Divisible by both 2 and 3                                                                                                         |
| Divisible by 11       | (Sum of odd-place digits) – (sum of even-place digits) = 0 or a multiple of 11                                                    |
| Unit digit cycles     | 2, 3, 7, 8 → cycle of 4 · 4, 9 → cycle of 2 · 0, 1, 5, 6 → always the same. Divide the power by 4; remainder 0 → use the 4th term |
| Number of factors     | $N = a^p \times b^q \times c^r \rightarrow (p + 1)(q + 1)(r + 1)$                                                                 |
| Sums                  | $1 + \dots + n = n(n + 1)/2$ · squares $n(n + 1)(2n + 1)/6$ · cubes $[n(n + 1)/2]^2$ · first n odd numbers = $n^2$                |
| Division              | Dividend = divisor × quotient + remainder                                                                                         |
| Trailing zeros in n!  | $[n/5] + [n/25] + [n/125] + \dots$                                                                                                |
| Multiples in a range  | Numbers from 1 to N divisible by k = $[N/k]$                                                                                      |

## Patterns — the method and one solved example each

| Pattern                    | Method                    | Solved example                                                                                                        |
|----------------------------|---------------------------|-----------------------------------------------------------------------------------------------------------------------|
| 1. Divisibility            | Apply the rule            | Is 7,58,452 divisible by 4? Last two digits 52 → <b>yes</b> . 1331 by 11? $1 - 3 + 3 - 1 = 0 \rightarrow$ <b>yes</b>  |
| 2. Missing digit           | Digit-sum rule            | 5_2 divisible by 9 → $5 + x + 2 = 9 \rightarrow x = 2$ (522 = $9 \times 58$ )                                         |
| 3. Unit digit of a power   | Cycle + remainder         | $7^{45}$ : cycle 7, 9, 3, 1; $45 \div 4$ leaves 1 → <b>7</b>                                                          |
| 4. Unit digit of a product | Multiply unit digits only | $23 \times 47 \times 58 \rightarrow 3 \times 7 = 21 \rightarrow 1 \times 8 \rightarrow$ <b>8</b>                      |
| 5. Remainder of a power    | Find the remainder cycle  | $2^{10} \div 7$ : remainders 2, 4, 1 repeat; $10 \div 3$ leaves 1 → <b>2</b> ( $1024 = 7 \times 146 + 2 \checkmark$ ) |
| 6. Number of factors       | Prime-factorise           | $72 = 2^3 \times 3^2 \rightarrow 4 \times 3 =$ <b>12</b>                                                              |
| 7. Trailing zeros          | Count the 5s              | $100! \rightarrow 20 + 4 =$ <b>24</b>                                                                                 |
| 8. Series sums             | Formulas                  | 1 to 50 = <b>1275</b> · squares 1-10 = <b>385</b> · cubes 1-10 = <b>3025</b>                                          |
| 9. Find the dividend       | $D \times Q + R$          | Divisor 17, quotient 9, remainder 5 → <b>158</b>                                                                      |
| 10. Counting multiples     | $[N/k]$                   | Numbers from 1 to 100 divisible by 3 → <b>33</b>                                                                      |

### Traps

- **1 is not a prime**, and **2 is the only even prime**.
- Unit digits: if the power is a multiple of 4, use the **4th** term of the cycle (e.g.  $3^4 \rightarrow 1$ ), not 'zero'.
- For 11, alternate the signs from the right or the left consistently — don't just add all the digits.

## B · REASONING — Coding-Decoding

A word is turned into a code by a hidden rule. **Method:** write the letter positions above both the word and the code, then find the difference letter by letter. Check, in this order: same shift (+n/−n), opposite letter (sum 27), reversed order, alternating shifts, shift by position (+1, +2, +3 ...).

| Coding type         | What to look for                                                                |
|---------------------|---------------------------------------------------------------------------------|
| Letter shift        | Every letter moves by the same number (CAT → DBU is +1)                         |
| Opposite letters    | Each pair of positions sums to 27 (A ↔ Z, G ↔ T)                                |
| Reversal            | Letters written backwards, sometimes with a shift added                         |
| Number coding       | Letters → positions; sometimes the code is a sum or product of positions        |
| Substitution        | 'Sky is called sea, sea is called water ...' — follow the chain carefully       |
| Sentence comparison | Find the word common to two coded sentences, then match it with the common code |

### Patterns, with solved examples

| Pattern             | Example                                                                 | Solution                                   |
|---------------------|-------------------------------------------------------------------------|--------------------------------------------|
| Same shift          | CAT → DBU. DOG → ?                                                      | +1 each → <b>EPH</b>                       |
| Opposite letters    | GOOD → TLLW. BAD → ?                                                    | B ↔ Y, A ↔ Z, D ↔ W → <b>YZW</b>           |
| Reversal            | PEN → NEP. BOOK → ?                                                     | <b>KOOB</b>                                |
| Alternating shift   | ROSE → SNTD. Rule?                                                      | +1, −1, +1, −1                             |
| Shift by position   | ABC → BDF. CAT → ?                                                      | +1, +2, +3 → <b>DCW</b>                    |
| Position sum        | CAT = 24. DOG = ?                                                       | 4 + 15 + 7 = <b>26</b>                     |
| Digits for letters  | BAD = 214. CAB = ?                                                      | <b>312</b>                                 |
| Substitution        | Sky → sea, sea → water, water → air. What do we drink?                  | We drink water, which is called <b>air</b> |
| Sentence comparison | 'pit na sa' = you are good; 'na ka sa' = they are good. Code for 'you'? | 'na sa' = 'are good' → <b>pit</b>          |

### Speed rule

Never guess the rule from the first letter alone. Confirm it on at least three letters before applying it.

# C · SCIENCE — Physics 5: Waves & Sound

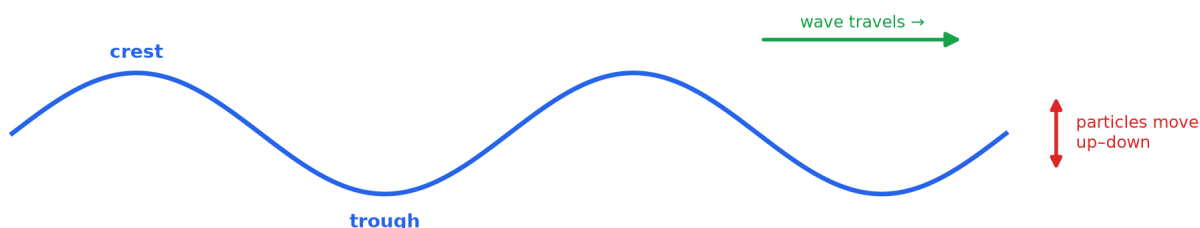
8 lessons. Sound is a favourite source of easy marks: a handful of numbers (20 Hz, 20 kHz, 17 m, 330–343 m/s) and a few clear ideas.

## Lesson 1 · What is a wave? (about 6 min)

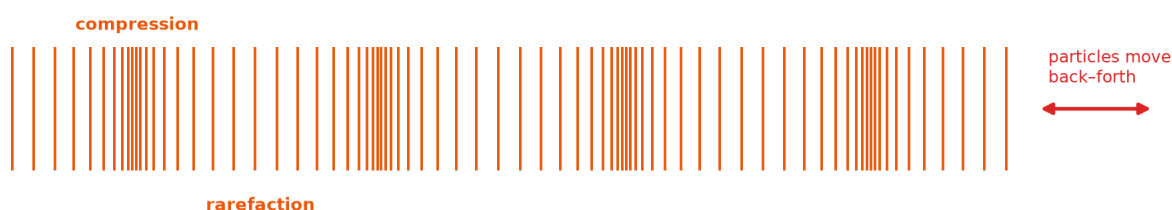
Drop a stone in a pond: ripples spread out, but a floating leaf only bobs up and down — it doesn't travel with the ripples. A **wave carries energy from one place to another without carrying the material**.

| Mechanical waves                                                                                   | Electromagnetic waves                                                                                  |
|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| <b>Need a medium</b> (solid, liquid or gas) — sound, water waves, waves on a string, seismic waves | <b>Need no medium</b> — travel through vacuum at $3 \times 10^8$ m/s: light, radio, microwaves, X-rays |

**TRANSVERSE:** particles vibrate  $\perp$  to the wave direction (water surface, string, light)



**LONGITUDINAL:** particles vibrate along the wave direction (sound in air)



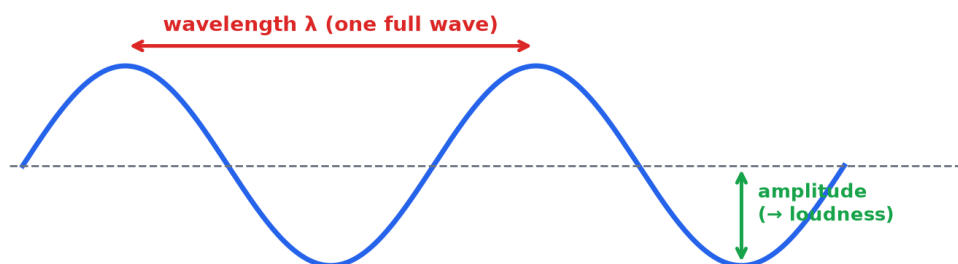
**Sound in air is longitudinal** (compressions and rarefactions). **Light is transverse**. Sound in solids can be both.

### Quick check 1 — answer out loud before moving on (answers at the end)

a) Is sound in air transverse or longitudinal? b) Can light travel through a vacuum? Can sound?

## Lesson 2 · Wavelength, frequency, amplitude and wave speed

(about 5 min)



frequency  $f$  = waves per second (Hz) → pitch · time period  $T = 1/f$  · speed  $v = f \times \lambda$

| Term                     | Meaning                                      | Unit       |
|--------------------------|----------------------------------------------|------------|
| Wavelength ( $\lambda$ ) | Length of one complete wave (crest to crest) | metre      |
| Frequency (f)            | Number of waves passing per second           | hertz (Hz) |
| Time period (T)          | Time for one wave; $T = 1/f$                 | second     |
| Amplitude                | Maximum disturbance from the rest position   | metre      |
| Wave speed (v)           | $v = f \times \lambda$                       | m/s        |

### Worked example — wave speed

A sound has frequency 500 Hz and wavelength 0.68 m:  $v = 500 \times 0.68 = \mathbf{340 \text{ m/s}}$ .

Time period of a 50 Hz wave:  $T = 1/50 = \mathbf{0.02 \text{ s}}$ .

### Quick check 2 — answer out loud before moving on (answers at the end)

A wave has frequency 200 Hz and wavelength 1.7 m. What is its speed?

## Lesson 3 · How sound is made and how fast it travels (about 6 min)

Sound is made by **vibrating objects**: vocal cords, a tuning fork, guitar strings, a drum skin. It **needs a medium**: a ringing bell in a jar becomes silent as the air is pumped out. That's why astronauts talk by radio in space, and there is no sound on the Moon.

| Medium             | Speed of sound (approx.)                                                               |
|--------------------|----------------------------------------------------------------------------------------|
| Air (0 °C / 20 °C) | $\approx 332 \text{ m/s}$ / $\approx 343 \text{ m/s}$ (questions often use 330 or 340) |
| Water              | $\approx 1500 \text{ m/s}$                                                             |
| Steel / iron       | $\approx 5000\text{--}6000 \text{ m/s}$                                                |

**Order: solids > liquids > gases.** In air, sound travels faster when it's **warmer** or more **humid**. At constant temperature, changing the air pressure does **not** change the speed.

### Worked example — thunder

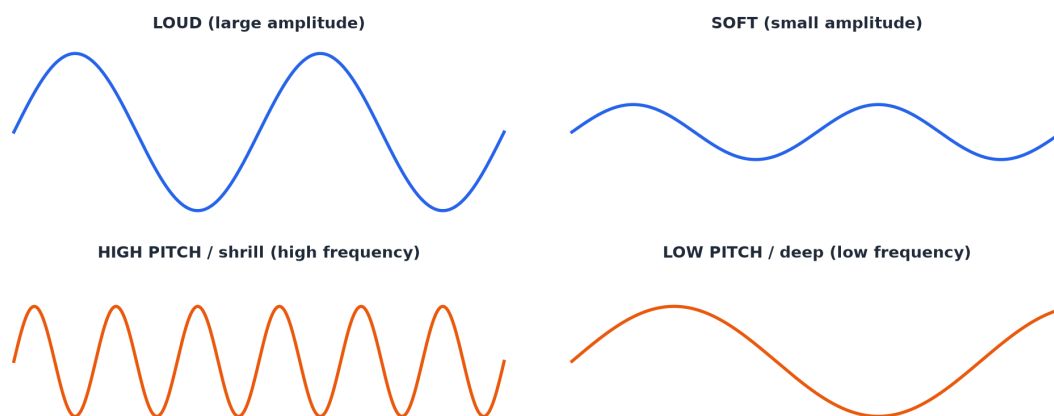
You see lightning and hear the thunder 3 s later. Light arrives almost instantly ( $3 \times 10^8 \text{ m/s}$ ).

Distance  $\approx 340 \times 3 = \mathbf{1020 \text{ m} \approx 1 \text{ km}}$ .

### Quick check 3 — answer out loud before moving on (answers at the end)

a) In which medium does sound travel fastest: air, water or steel? b) Why do we see lightning before we hear thunder?

## Lesson 4 · Loudness, pitch and quality (about 5 min)



| Feature                   | Depends on                                          | Examples                                                                                                                         |
|---------------------------|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>Loudness</b>           | <b>Amplitude</b> ; measured in <b>decibels (dB)</b> | Whisper $\approx$ 30 dB · conversation $\approx$ 60 dB · long exposure above $\approx$ 85 dB can damage hearing                  |
| <b>Pitch</b> (shrillness) | <b>Frequency</b>                                    | Women's and children's voices have higher pitch than men's · a mosquito is high-pitched, a lion's roar is low-pitched (but loud) |
| <b>Quality / timbre</b>   | Shape of the wave (mix of frequencies)              | Lets you tell a flute from a violin playing the same note, or recognise a friend's voice                                         |

### Exam trap

A lion's roar is **loud but low-pitched**; a mosquito's buzz is **soft but high-pitched**. Loudness and pitch are independent.

### Quick check 4 — answer out loud before moving on (answers at the end)

a) Pitch depends on what? b) What is the unit of loudness?

## Lesson 5 · Audible, infrasonic and ultrasonic sound (about 5 min)

| 20 Hz                          |                  | 20,000 Hz (20 kHz)                      |  |
|--------------------------------|------------------|-----------------------------------------|--|
| INFRASONIC                     | AUDIBLE (humans) | ULTRASONIC                              |  |
| earthquakes, whales, elephants | speech, music    | bats, dolphins, SONAR, ultrasound scans |  |

Humans hear roughly **20 Hz to 20,000 Hz (20 kHz)**. Below that is **infrasonic**; above it is **ultrasonic**. Dogs can hear some ultrasonic sounds (hence 'silent' dog whistles).

| Uses of ultrasound                        | How                                                        |
|-------------------------------------------|------------------------------------------------------------|
| Bats and dolphins navigate (echolocation) | Send ultrasound, listen for echoes                         |
| SONAR — Sound Navigation And Ranging      | Measures sea depth and finds submarines and shoals of fish |
| Ultrasonography (scans)                   | Images of organs and of a baby in the womb                 |
| Industry                                  | Detects cracks in metal, cleans hard-to-reach parts        |
| Medicine                                  | Breaks up kidney stones (lithotripsy)                      |

### Exam trap

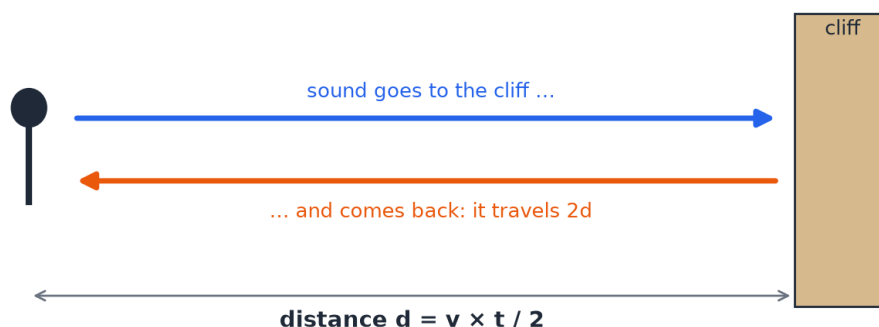
**Supersonic** means a **speed** greater than the speed of sound (jets, bullets) — it's not about frequency. Breaking the sound barrier makes a **sonic boom**. Don't confuse it with ultrasonic.

### Quick check 5 — answer out loud before moving on (answers at the end)

a) What is the audible range for humans? b) What does 'supersonic' describe?

## Lesson 6 · Echo, reverberation and SONAR (about 6 min)

An **echo** is reflected sound heard separately from the original. Our ear keeps a sound for about **0.1 s** (persistence of hearing), so the reflection must arrive at least 0.1 s later.



### Worked example — echo distances

**Minimum distance for an echo** ( $v = 340 \text{ m/s}$ ): sound travels  $340 \times 0.1 = 34 \text{ m}$  there and back →  $d = 17 \text{ m}$ .

Echo heard after 2 s from a cliff:  $d = 340 \times 2 \div 2 = 340 \text{ m}$ .

SONAR pulse returns from the sea bed after 2 s ( $v$  in water =  $1500 \text{ m/s}$ ): depth =  $1500 \times 2 \div 2 = 1500 \text{ m}$ .

**Reverberation** = many quick reflections blending together in a big hall. It's reduced with curtains, carpets and soft boards. Uses of reflection of sound: **stethoscope**, **megaphone**, curved ceilings and sound boards in auditoriums.

### Exam trap

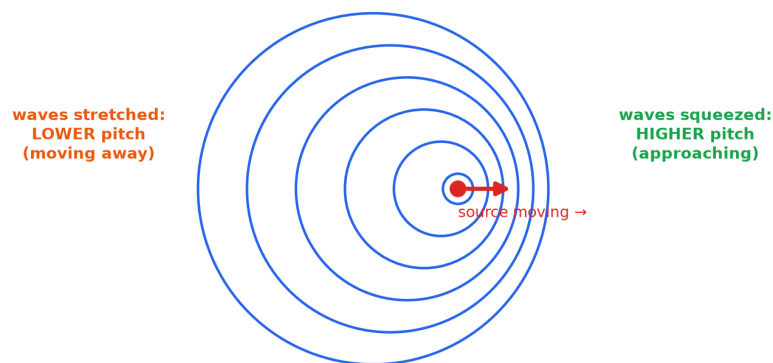
Always **divide by 2** in echo and SONAR questions — the sound goes there **and back**.

### Quick check 6 — answer out loud before moving on (answers at the end)

An echo is heard 3 s after a shout ( $v = 340 \text{ m/s}$ ). How far away is the wall?

## Lesson 7 · Doppler effect (about 3 min)

When a train approaches, its whistle sounds **higher**; as it moves away, **lower**. The source's real frequency doesn't change — the **heard** frequency does, because the waves get squeezed ahead of a moving source and stretched behind it.



Uses: police **radar speed guns**, Doppler ultrasound for blood flow, and astronomy (light from galaxies moving away is shifted towards red: **redshift**).

### Quick check 7 — answer out loud before moving on (answers at the end)

How does the pitch of an ambulance siren seem to change as it passes you?

## Lesson 8 · Resonance and the human ear (about 4 min)

Every object has a **natural frequency**. If it's pushed at exactly that frequency, it vibrates with a very large amplitude: **resonance**. Soldiers **break step** while crossing a bridge so their marching can't match the bridge's natural frequency. Tuning a radio to a station is (electrical) resonance.

| Part of the ear | Key facts                                                                                                                               |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Outer ear       | Collects sound; the eardrum vibrates                                                                                                    |
| Middle ear      | Three tiny bones — hammer (malleus), anvil (incus), stirrup ( <b>stapes</b> ). The stapes is the <b>smallest bone</b> in the human body |
| Inner ear       | <b>Cochlea</b> turns vibrations into nerve signals sent to the brain                                                                    |

### Quick check 8 — answer out loud before moving on (answers at the end)

a) Why do soldiers break step on a bridge? b) What is the smallest bone in the human body?



# Science 1-page summary — revise from this tomorrow

| Topic           | Must remember                                                                                                                                                |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Waves           | Carry energy, not matter · mechanical need a medium · EM (light, radio, X-ray) don't                                                                         |
| Types           | Transverse: crests/troughs (light, water surface) · longitudinal: compressions/rarefactions (sound in air)                                                   |
| Terms           | $v = f\lambda$ · $T = 1/f$ · amplitude → loudness · frequency → pitch                                                                                        |
| Speed of sound  | Solid > liquid > gas · air $\approx 330$ -343 m/s · water $\approx 1500$ · steel $\approx 5000$ + · rises with temperature and humidity · no sound in vacuum |
| Characteristics | Loudness = amplitude (dB) · pitch = frequency · quality = waveform                                                                                           |
| Ranges          | Audible 20 Hz-20 kHz · infrasonic below (earthquakes, whales, elephants) · ultrasonic above (bats, SONAR, scans)                                             |
| Supersonic      | SPEED faster than sound · sonic boom                                                                                                                         |
| Echo            | Needs $\geq 0.1$ s → minimum 17 m · $d = vt/2$ · SONAR depth = $vt/2$ · reverberation reduced by soft materials                                              |
| Doppler         | Approaching → higher pitch · receding → lower · radar guns, redshift                                                                                         |
| Resonance       | Forced = natural frequency → large amplitude · soldiers break step · radio tuning                                                                            |
| Ear             | Malleus, incus, stapes (smallest bone) · cochlea                                                                                                             |

## Science practice set — 20 questions · target 6 minutes

| #  | Question                                                                                                                                                                                            |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Sound cannot travel through: (a) water (b) steel (c) air (d) a vacuum                                                                                                                               |
| 2  | Sound travels fastest in: (a) water (b) air (c) vacuum (d) steel                                                                                                                                    |
| 3  | Sound waves in air are: (a) transverse (b) electromagnetic (c) both transverse and electromagnetic (d) longitudinal                                                                                 |
| 4  | Light waves are: (a) longitudinal (b) mechanical (c) transverse (d) sound waves                                                                                                                     |
| 5  | The pitch of a sound depends on its: (a) amplitude (b) speed (c) frequency (d) wavelength only                                                                                                      |
| 6  | The loudness of a sound depends on its: (a) frequency (b) amplitude (c) pitch (d) time period                                                                                                       |
| 7  | The unit used to measure sound level is the: (a) hertz (b) watt (c) pascal (d) decibel                                                                                                              |
| 8  | The audible range of human hearing is about: (a) 20 Hz to 20 kHz (b) 2 Hz to 2 kHz (c) 20 kHz to 200 kHz (d) 0 Hz to 20 Hz                                                                          |
| 9  | Bats find their way in the dark using: (a) infrasonic waves (b) light waves (c) ultrasonic waves (d) radio waves                                                                                    |
| 10 | SONAR uses: (a) infrasonic waves (b) microwaves (c) ultrasonic waves (d) X-rays                                                                                                                     |
| 11 | Sound of frequency below 20 Hz is called: (a) ultrasonic (b) infrasonic (c) supersonic (d) audible                                                                                                  |
| 12 | 'Supersonic' refers to: (a) a speed greater than the speed of sound (b) a frequency above 20 kHz (c) a frequency below 20 Hz (d) very loud sound                                                    |
| 13 | The minimum distance needed to hear an echo (speed of sound 340 m/s) is: (a) 17 m (b) 34 m (c) 10 m (d) 340 m                                                                                       |
| 14 | An echo is heard 2 s after a shout ( $v = 340$ m/s). The distance of the reflecting cliff is: (a) 680 m (b) 170 m (c) 340 m (d) 1360 m                                                              |
| 15 | A wave of frequency 200 Hz has wavelength 1.7 m. Its speed is: (a) 340 m/s (b) 117.6 m/s (c) 201.7 m/s (d) 34 m/s                                                                                   |
| 16 | The whistle of an approaching train sounds higher because of: (a) resonance (b) echo (c) the Doppler effect (d) reverberation                                                                       |
| 17 | Soldiers break step while crossing a bridge to avoid: (a) echo (b) the Doppler effect (c) reverberation (d) resonance                                                                               |
| 18 | The smallest bone in the human body is the: (a) stapes (b) femur (c) malleus (d) incus                                                                                                              |
| 19 | Women's voices usually have a higher pitch than men's because they have a higher: (a) amplitude (b) loudness (c) frequency (d) wavelength                                                           |
| 20 | We see lightning before we hear thunder because: (a) sound travels faster than light (b) light travels much faster than sound (c) thunder is produced later (d) our eyes react faster than our ears |

| Score | Time taken | Wrong Q numbers | Mistake type (concept / formula / calc / careless / recall) |
|-------|------------|-----------------|-------------------------------------------------------------|
| / 20  |            |                 |                                                             |

# Answers

## Science practice set

| #  | Ans | Why / lesson to re-read            |
|----|-----|------------------------------------|
| 1  | d   | Lesson 3                           |
| 2  | d   | Solid > liquid > gas (Lesson 3)    |
| 3  | d   | Lesson 1                           |
| 4  | c   | Lesson 1                           |
| 5  | c   | Lesson 4                           |
| 6  | b   | Lesson 4                           |
| 7  | d   | Lesson 4                           |
| 8  | a   | Lesson 5                           |
| 9  | c   | Lesson 5                           |
| 10 | c   | Lesson 5                           |
| 11 | b   | Lesson 5                           |
| 12 | a   | Lesson 5 trap                      |
| 13 | a   | $340 \times 0.1 \div 2$ (Lesson 6) |
| 14 | c   | $v\lambda/2$ (Lesson 6)            |
| 15 | a   | $v = f\lambda$ (Lesson 2)          |
| 16 | c   | Lesson 7                           |
| 17 | d   | Lesson 8                           |
| 18 | a   | Lesson 8                           |
| 19 | c   | Lesson 4                           |
| 20 | b   | Lesson 3                           |

## Quick checks

| QC | Answers                                                                                               |
|----|-------------------------------------------------------------------------------------------------------|
| 1  | a) longitudinal b) light can; sound cannot                                                            |
| 2  | $v = 200 \times 1.7 = 340 \text{ m/s}$                                                                |
| 3  | a) steel b) light ( $3 \times 10^8 \text{ m/s}$ ) is far faster than sound ( $\sim 340 \text{ m/s}$ ) |
| 4  | a) frequency b) decibel (dB)                                                                          |
| 5  | a) about 20 Hz to 20 kHz b) a speed greater than the speed of sound                                   |
| 6  | $d = 340 \times 3 \div 2 = 510 \text{ m}$                                                             |
| 7  | Higher as it approaches, then lower as it moves away                                                  |
| 8  | a) to avoid resonance with the bridge's natural frequency b) the stapes (stirrup), in the middle ear  |

## Day 4 recall (Block 0)

| # | Answer                               | #  | Answer                          |
|---|--------------------------------------|----|---------------------------------|
| 1 | Pascal                               | 6  | Bernoulli's principle           |
| 2 | Liquid pressure increases with depth | 7  | Steel                           |
| 3 | Pascal's law                         | 8  | Radiation                       |
| 4 | Sea water is denser                  | 9  | 4 °C                            |
| 5 | Surface tension                      | 10 | $10,000 \times (1/10)^2 = ₹100$ |

### Send me tonight

Day 4 recall score /10 · science practice score /20 and time · Maths and Reasoning: attempted, correct, average seconds per question · the question numbers you got wrong.